CALCULUS II - AY2024-25

Chapter 1. Functions of Several Variables

Outline

1. Summary
2. Vector space $\mathbb{R}^n$
3. Limits
4. Continuity
5. Partial derivatives
6. The Chain Rules
7. Approximations

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Summary

- We extend the basic ideas of single-variable calculus to functions of several variables.
- We discuss limit at a point and continuity.
- As there are more than one variables, the derivatives involved are more interesting. Applications of derivatives are also discussed.
- Similarly, we also discuss integrals involving several variables.

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• The set $\mathbb{R}^n$

The set $\mathbb{R}^{\mathbf{n}}$ consists of all n-tuples $(x_1, x_2, \ldots, x_n)$ with real entries.

The addition and scalar multiplication operations are coordinate-wise, i.e.

1. $(x_1, x_2, \ldots, x_n) + (y_1, y_2, \ldots, y_n) = (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n)$
2. $c(x_1, x_2, \ldots, x_n) = (cx_1, cx_2, \ldots, cx_n), \quad c \in \mathbb{R}.$

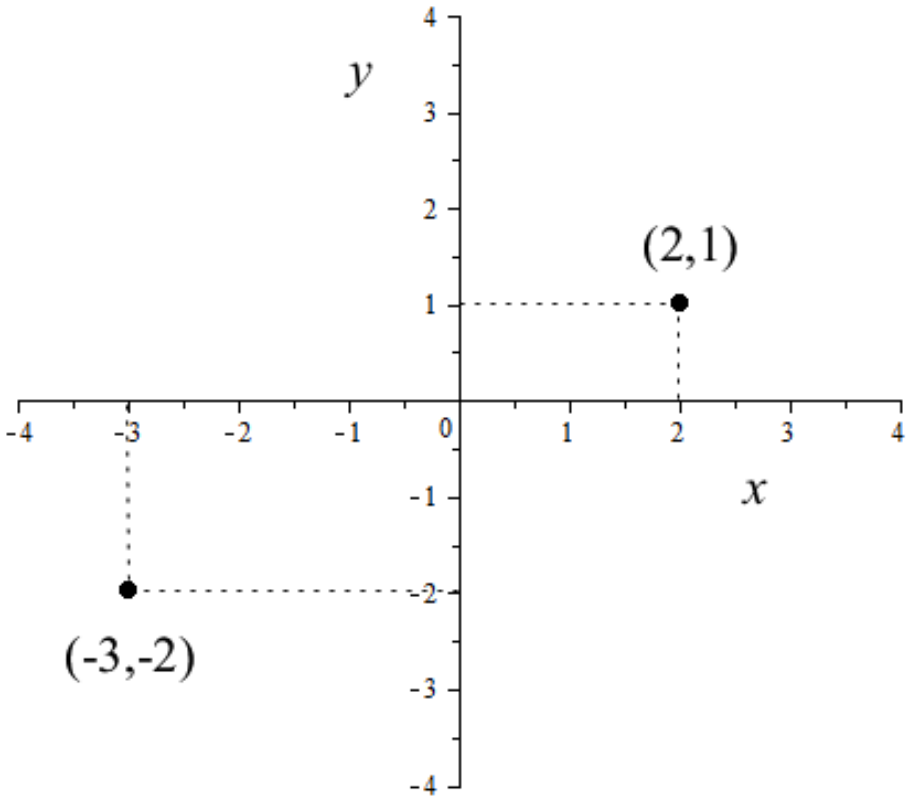
Then $\mathbb{R}^n$ becomes a vector space, with the standard basis

$$
\begin{aligned}
e_1 &= (1, 0, 0, \ldots, 0), \\
e_2 &= (0, 1, 0, \ldots, 0), \\
\ldots \quad &\ldots \quad \ldots \quad \ldots \\
e_n &= (0, 0, 0, \ldots, 1).
\end{aligned}
$$

In particular,

- When $n = 1$, we have $\mathbb{R}^1 = \mathbb{R}$, the usual real line.
- When $n = 2$, $\mathbb{R}^2$ is the 2-dimensional plane (we usually call it the xy-plane and write (x, y) instead of (x_1, x_2) for points in $\mathbb{R}^2$).
- When $n = 3$, $\mathbb{R}^3$ is the 3-dimensional space we live in (we usually call it the xyz-plane and write (x, y, z) instead of (x_1, x_2, x_3) for points in $\mathbb{R}^3$).

• The xy-plane $\mathbb{R}^2$



Lines, circles & ellipses in $\mathbb{R}^2$

- Equation of a line:

$$ax + by = c, \quad (a^2 + b^2 \neq 0).$$

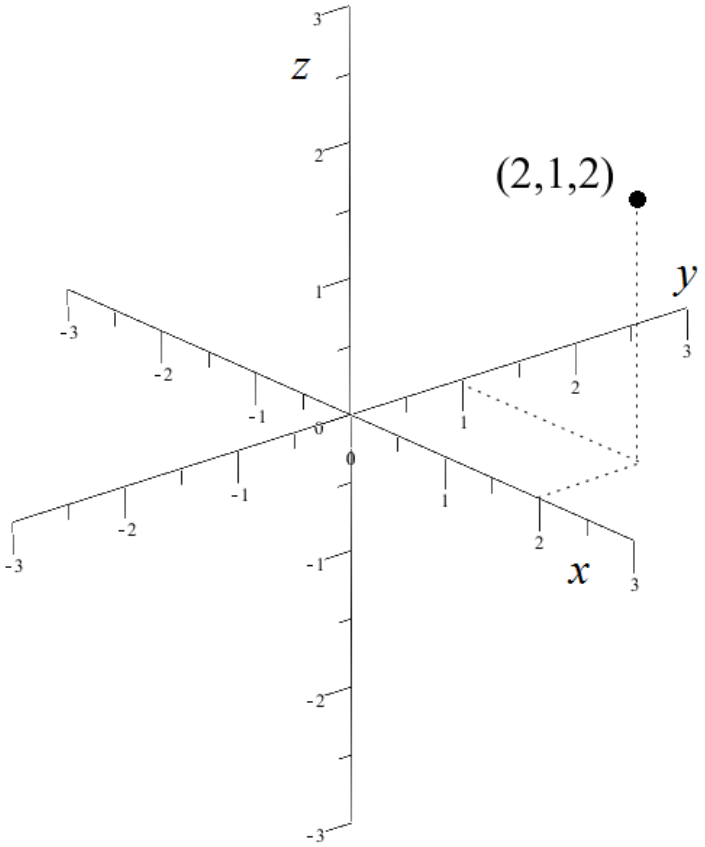
- Equation of a circle:

$$(x - a)^2 + (y - b)^2 = r^2, \quad r > 0.$$

- Equation of an ellipse:

$$\alpha(x - a)^2 + \beta(y - b)^2 = r^2, \quad \alpha > 0, \beta > 0, r > 0.$$

• The *xyz*-space $\mathbb{R}^3$



Note. Use the right-hand rule to determine the positive z-axis.

• The Swiss 200-francs banknote



Planes in $\mathbb{R}^3$

- Equation of a plane in $\mathbb{R}^3$:

$$ax + by + cz = d, \quad (a^2 + b^2 + c^2 \neq 0).$$

(Coefficients give a normal vector $\vec{n} = (a, b, c)$ to the plane.)

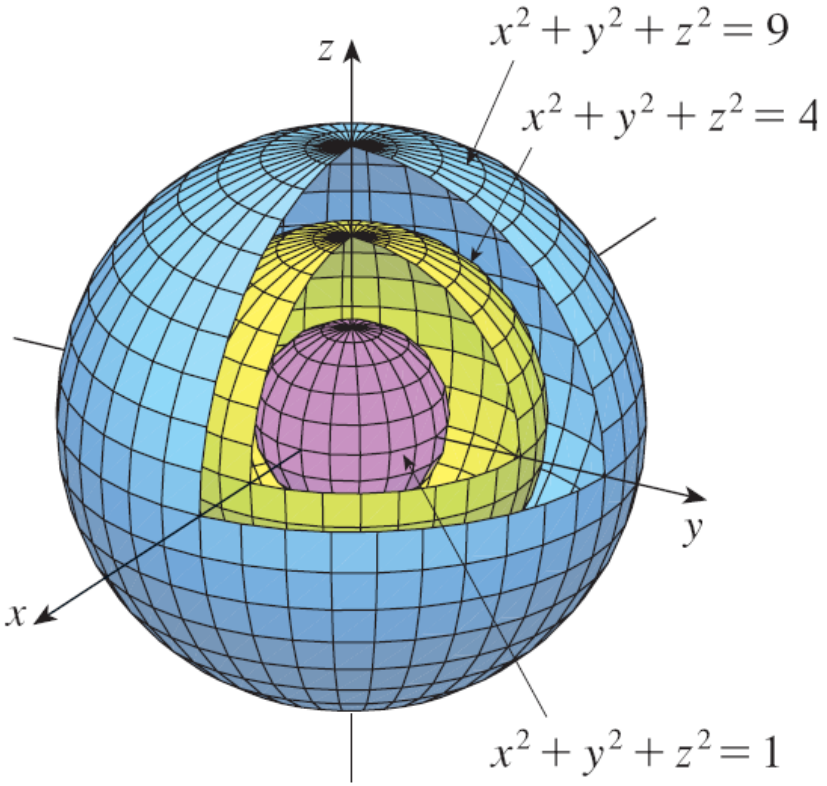
- Vector form:

$$(\vec{r} - \vec{r_0}) \cdot \vec{n} = 0.$$

Spheres in $\mathbb{R}^3$

- Equation of a sphere: center (a, b, c), radius r

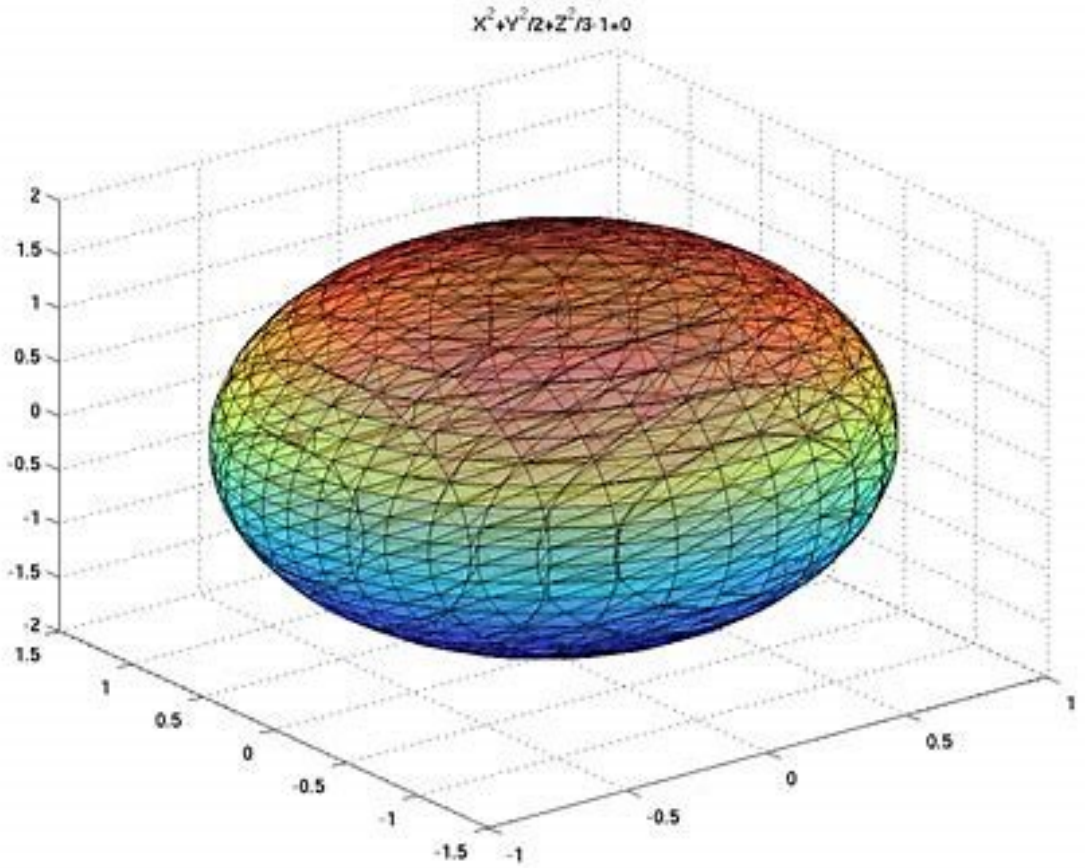
$$(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2, \quad r > 0.$$



Ellipsoids in $\mathbb{R}^3$

- Equation of an ellipsoid:

$$\alpha(x-a)^2+\beta(y-b)^2+\gamma(z-c)^2=r^2, \quad \alpha>0, \beta>0, \gamma>0, r>0.$$

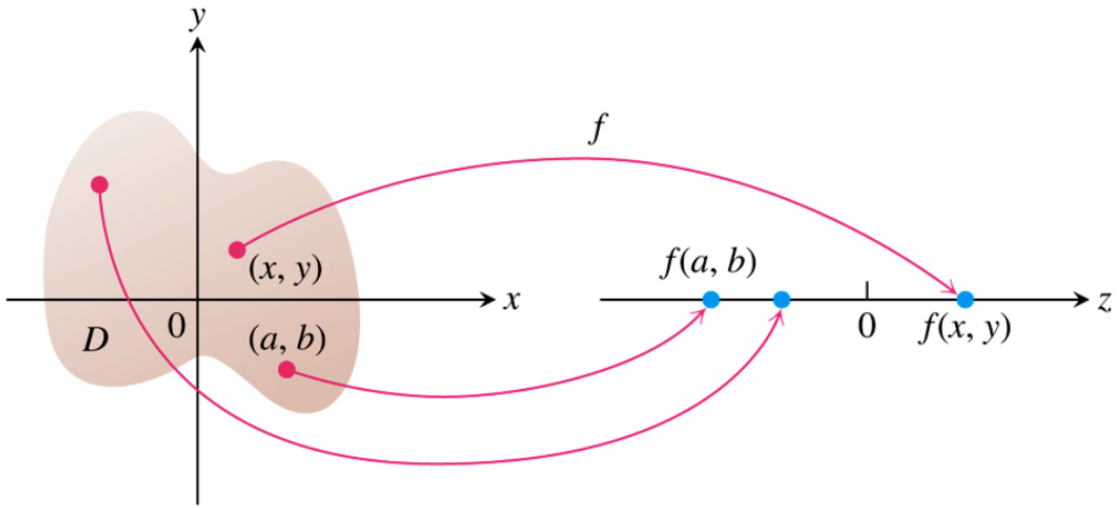


• Illustration of the ellipsoid in the real life



• Function of 2 variables

A **function** f of two variables, with a domain $D \subset \mathbb{R}^2$, is a rule that assigns to each point $(x, y) \in D$ a unique real number which is denoted by $f(x, y)$.



- We often write $z = f(x, y)$ to make explicit the value taken on by f at the point (x, y).
- The variables x and y are independent, and z is dependent.

Note. When the domain of f is not specified, we take the largest subset of $\mathbb{R}^2$ on which $f(x, y)$ is a real number, to be the domain of f.

Many functions depend on more than one independent variables.

Example

1. The power P (watts) of an electric circuit is related to the circuit's resistance R (ohms) and current I (amperes) by the equation

$$P = RI^2.$$

This means that P is a function of two variables R and I; we may express this function as $P(R, I) = RI^2$.

2. The distance of a point $P(x, y)$ from a point $Q(a, b)$ is given by

$$d(x, y) = \sqrt{(x - a)^2 + (y - b)^2}.$$

The domain of $d(x, y)$ is $\mathbb{R}^2$ since $d(x, y)$ is defined for every point in $\mathbb{R}^2$.

Example

Find and sketch the domain of $f(x,y) = \sqrt{y - x^2}$ and evaluate $f(1, \sqrt{2})$.

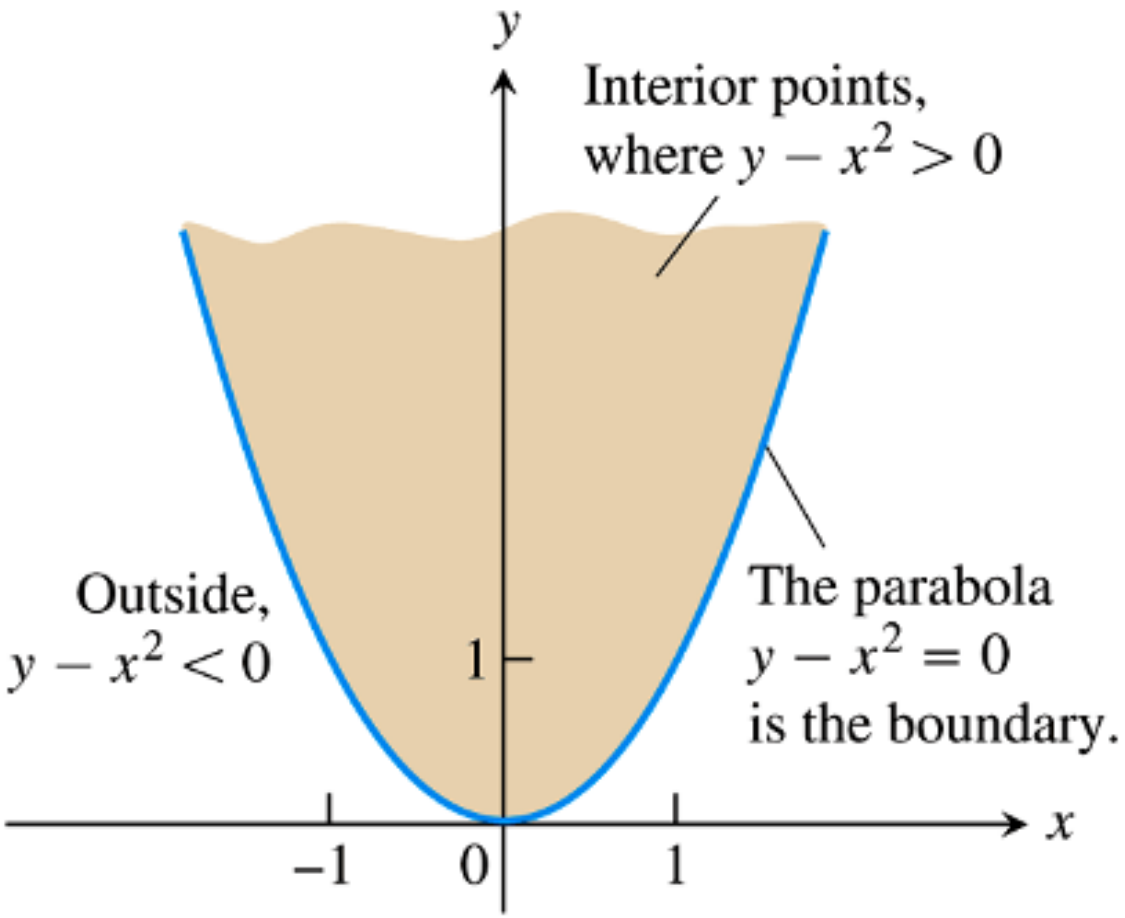
Solution. Note that $f(x,y)$ is defined whenever $y - x^2 \geq 0$. Then the domain of f is

$$D = \left\{(x,y) : y \geq x^2\right\}$$

and

$$f(1, \sqrt{2}) = \sqrt{\sqrt{2} - 1}.$$

- Sketch the domain D of f (**!**)


y
Interior points,
where $y - x^2 > 0$
Outside,
$y - x^2 < 0$
The parabola
$y - x^2 = 0$
is the boundary.
1
x
−1
0
1

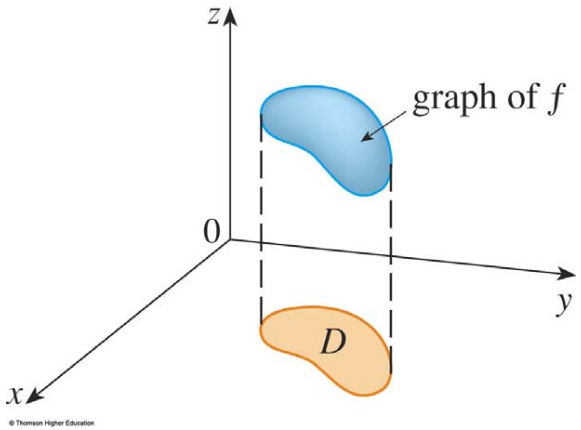
• Graph of $f(x, y)$

The graph of a function $z = f(x, y)$ is a 3-dimensional set

$$\left\{(x, y, z) \in \mathbb{R}^3 : z = f(x, y),\ (x, y) \in D\right\}.$$

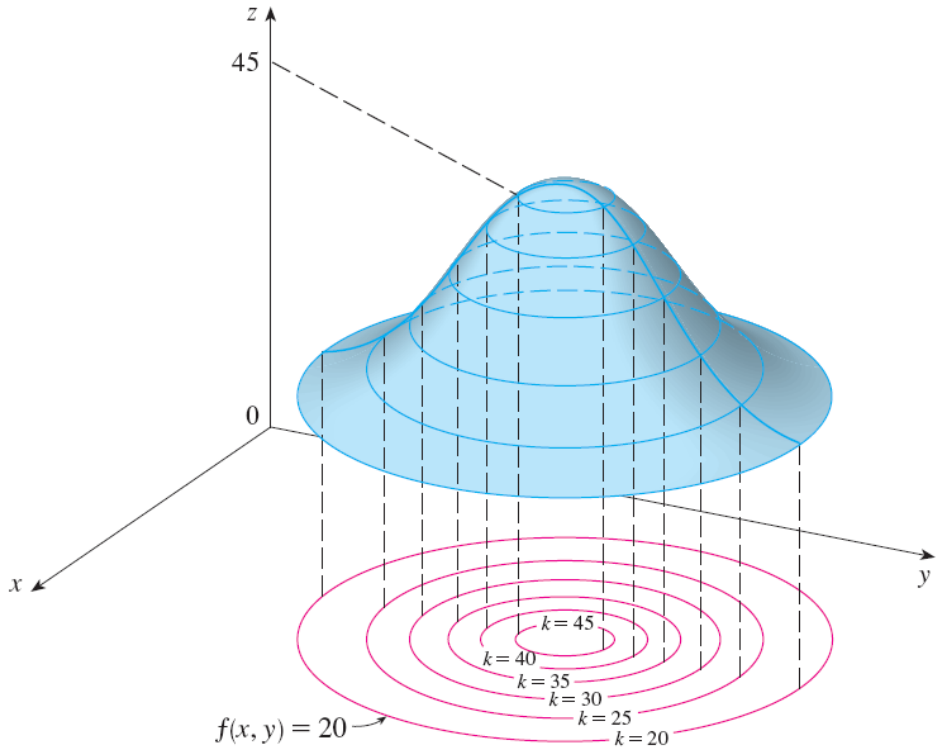
The graph of f is also called the surface $z = f(x, y)$.

It is a challenge to sketch it on paper.



• Level curves of $f(x, y)$

The set of points $(x, y) \in \mathbb{R}^2$ where $f(x, y)$ has the same (constant) value $f(x, y) = k$ is called a level curve of f.

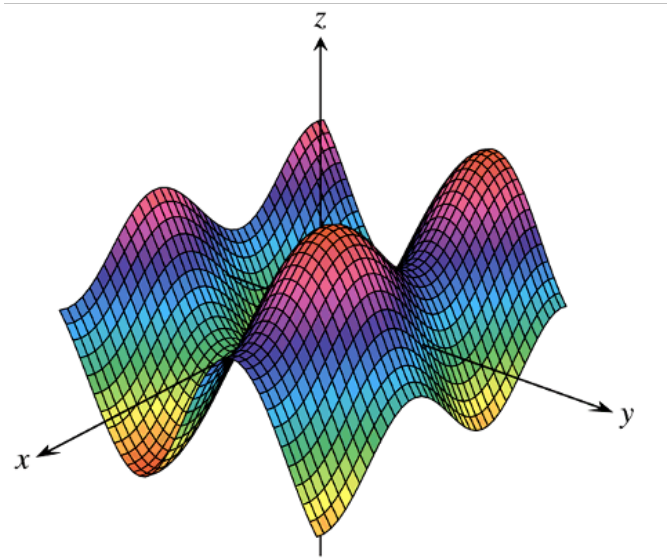


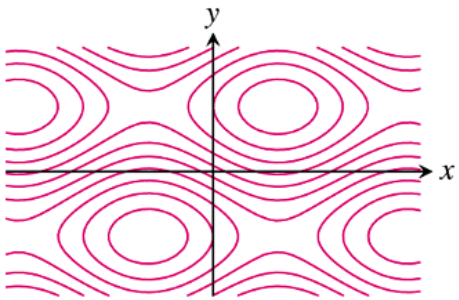
The key point is that a level curve $f(x, y) = k$ can be thought of as a horizontal slice of the graph at height $z = k$.

This slice is the intersection of the graph with the plane $z = k$.

$\Longrightarrow$ *An illustration of a level curve from "Math Insight"*

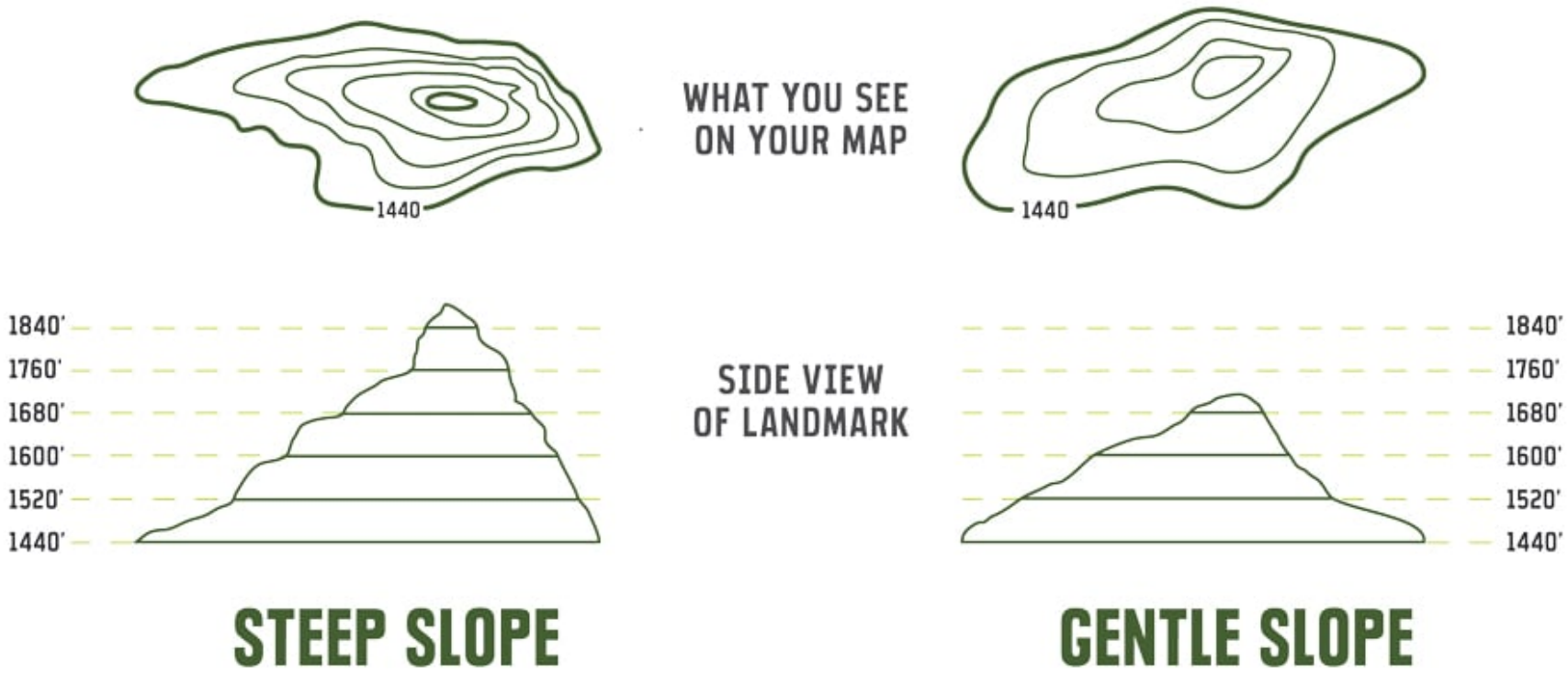
Below is the computer-generated graph and level curve of the function $f(x, y) = \sin x + 2 \sin y$.





• Level curves and Contour map

If $f(x, y)$ is the altitude function of a region in $\mathbb{R}^2$, then the level curves give rise to the contour map.



Example

Find the level curve of $f(x, y) = 100 - x^2 - y^2$, which

1. corresponds to $f(x, y) = 75$.
2. passes through $(10, 0)$.

Note. Our aim is to determine all points (x, y) lying on each of the above level curves (by describing (x, y) via equations and sketching them, if possible.)

Solution.

1. To find the level curve corresponds to $f(x, y) = 75$, we find (x, y) satisfying

$$100 - x^2 - y^2 = 75 \quad \Longleftrightarrow \quad x^2 + y^2 = 25.$$

This is a circle centered at $(0, 0)$ of a radius 5.

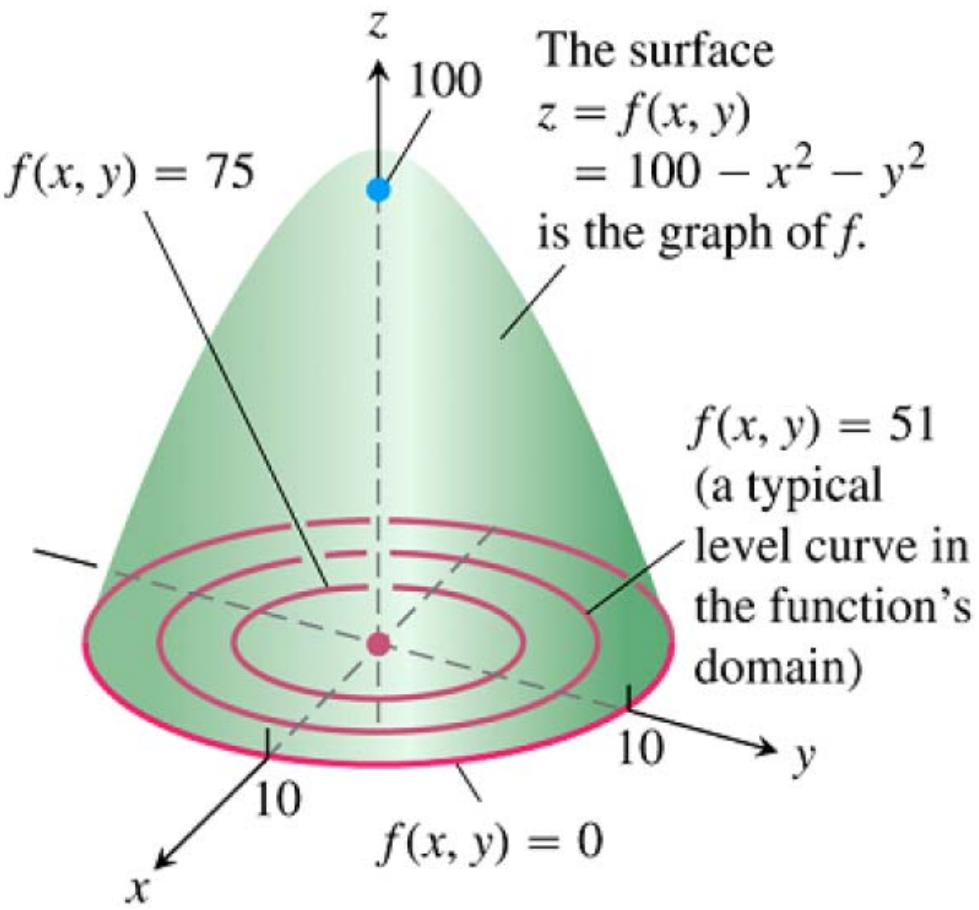
2. The level curve which passes through $(10, 0)$ is the level curve containing all points (x, y) corresponds to same value of f evaluated at $(10, 0)$, i.e., $f(10, 0)$.

Since $f(10, 0) = 0$, the required level curve is described by

$$f(x, y) = 0 \quad \Longleftrightarrow \quad x^2 + y^2 = 10^2.$$

This is a circle centered at $(0, 0)$ of a radius 10.

The graph & level curves of $f(x, y) = 100 - x^2 - y^2$:



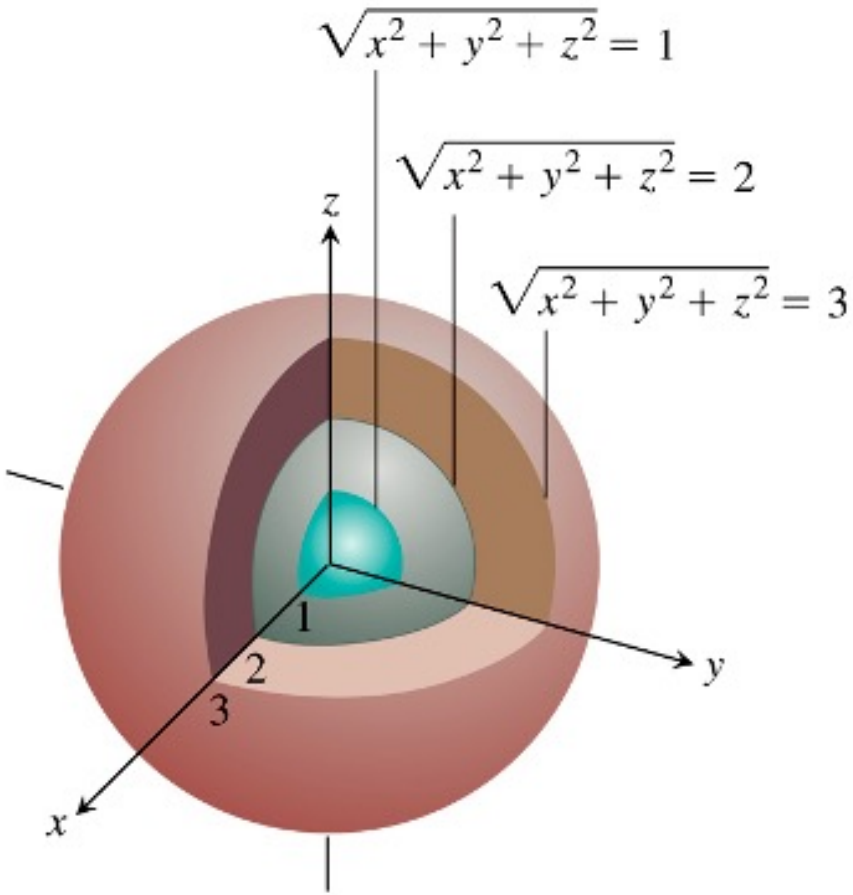
• Function of 3 variables

A function f of three variables, with a domain $D \subset \mathbb{R}^3$, is a rule that assigns to each point $(x, y, z) \in D$ a unique real number which is denoted by $f(x, y, z)$.

- We often write $w = f(x, y, z)$ to make explicit the value taken on by f at the point (x, y, z).
- The variables x, y, and z are independent, and w is dependent.

• Level surfaces of $f(x, y, z)$

For a function f of three variables, the set of points $(x, y, z) \in \mathbb{R}^3$ where $f(x, y, z)$ has a constant value $f(x, y, z) = k$ is called a level surface of f.



More precisely, for a function of three variables $f(x, y, z)$, we would need four dimensions to draw its graph.

The graph is the set of points $(x, y, z, f(x, y, z))$.

However, you may have a hard time visualizing what this graph in four dimensions would look like.

Instead, for a function of three variables, we can look at the level surface in three-dimensional space.

$\Longrightarrow$ *An illustration of a level surface from "Math Insight"*

Example

Consider the function $f(x, y, z) = (x-1)^2 + y^2 + z^2$.

1. Find the level surface which corresponds to $f(x, y, z) = 9$.
2. Find the level surface which passes through the point $(1, -2, 3)$.

Solution.

1. The level surface of f which corresponds to $f(x, y, z) = 9$, is

$$(x-1)^2 + y^2 + z^2 = 9.$$

This is the sphere centered at $(1, 0, 0)$ of radius 3.

2. Since $f(1, -2, 3) = 0^2 + (-2)^2 + 3^2 = 13$, the level surface of f that passes through $(1, -2, 3)$ is the set

$$\{(x, y, z) \in \mathbb{R}^3 : (x-1)^2 + y^2 + z^2 = 13\}.$$

This is the sphere centered at $(1, 0, 0)$ of radius $\sqrt{13}$.

• Functions of several variables

We can define functions of several variables and their level sets.

- **Functions of several variables.**

> A function f of n variables with a domain $D \subset \mathbb{R}^n$, is a rule that assigns to each point $(x_1, x_2, \ldots, x_n) \in D$ a unique real number denoted by $f(x_1, x_2, \ldots, x_n)$.

The variables $x_1, x_2, \ldots, x_n$ are independent variables, and the variable w is a dependent variable.

- **Level sets of functions of several variables.**

> For a function f of n variables, the set of points $(x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$ where $f(x_1, x_2, \ldots, x_n)$ has a constant value $f(x_1, x_2, \ldots, x_n) = k$ is called a level set of f.

The real number $w = f(x_1, x_2, \ldots, x_n)$ is called the evaluation of f at the point $(x_1, x_2, \ldots, x_n)$.

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- Recall (Calculus I): Intuitive concept.

For a function $f(x)$ of one variable x, intuitively, we say the limit of a function f when x tends to a point x_0 is L, if $f(x)$ gets very near to L when x gets very near to x_0.

Here,

- x gets very near to x_0 means that the distance $|x - x_0|$ becomes very small.
- $f(x)$ gets very near to L means that the distance $|f(x) - L|$ becomes very small.

- Recall (Calculus I): Formal definition.

Let f be a function of one variable defined on a domain $D \subset \mathbb{R}$, $x_0 \in \mathbb{R}$ (x_0 not necessary in D). We say that the limit of $f(x)$ as x tends to x_0 is L if

for any $\varepsilon > 0$ there exists a corresponding $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \quad \text{whenever} \quad 0 < |x - x_0| < \delta,\ x \in D,$$

and we write

$$\lim_{x \to x_0} f(x) = L.$$

• Limits in several variables: Intuitive concept

Now for function $f(x, y)$ of two variables x and y, intuitively, we say the limit of a function f when (x, y) tends to a point (x_0, y_0) is L, if $f(x, y)$ gets very near to L when (x, y) gets very near to (x_0, y_0) (in any possible way).

Remark.

(x, y) *gets very near to* (x_0, y_0) means that the *distance* between points (x, y) and (x_0, y_0) gets very small. This distance above is defined by

$$\|(x, y) - (x_0, y_0)\| := \sqrt{(x - x_0)^2 + (y - y_0)^2}.$$

• Limits

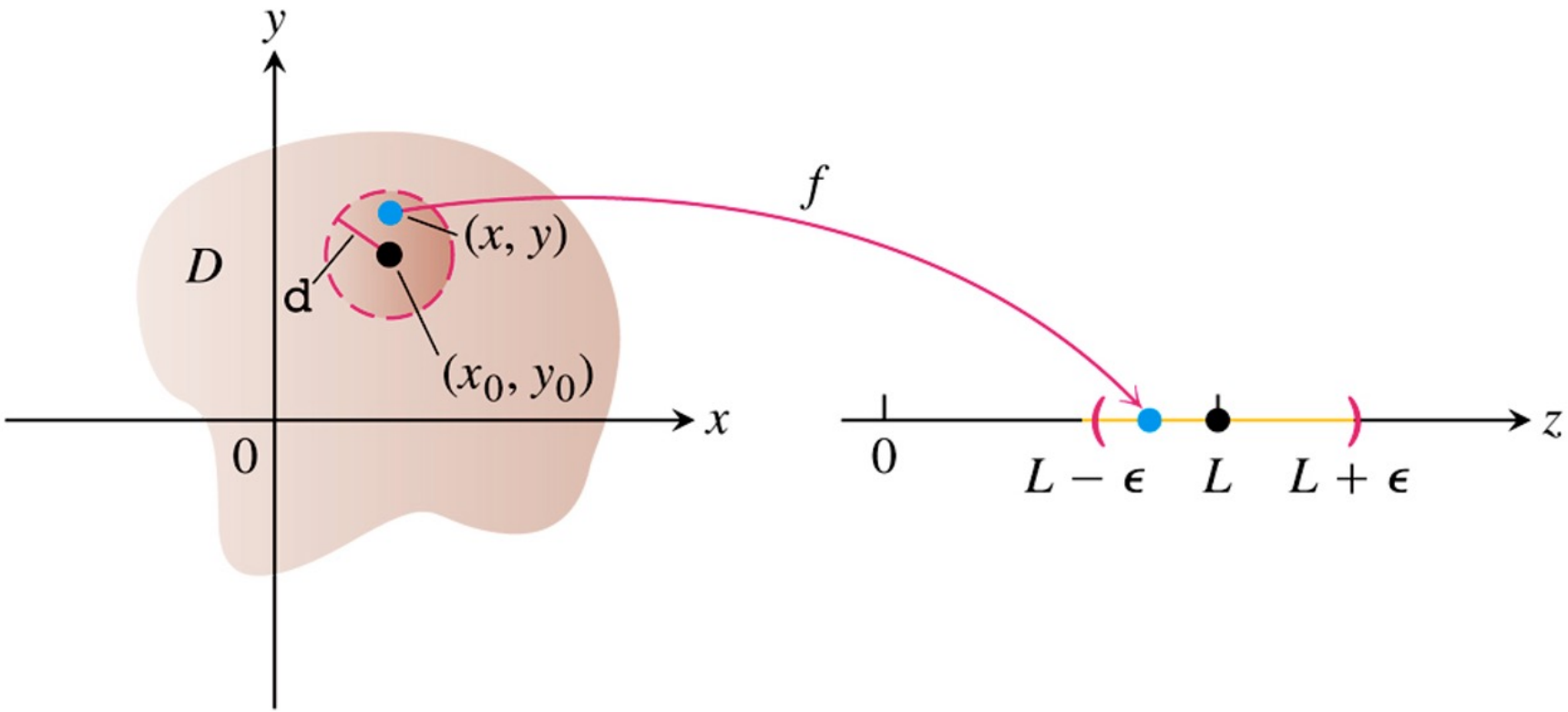
Let f be a function of two variables defined on a domain $D \subset \mathbb{R}^2$ and $(x_0, y_0) \in \mathbb{R}^2$ ((x_0, y_0) is not necessary in D).

We say that the limit of $f(x, y)$ as (x, y) tends to (x_0, y_0) exists, if there is a number L such that

> for any $\varepsilon > 0$ there exists a corresponding $\delta > 0$ such that
>
> $|f(x, y) - L| < \varepsilon$ whenever $0 < \|(x, y) - (x_0, y_0)\| < \delta,\ (x, y) \in D,$

and we write

$$\lim_{(x,y)\to(x_0,y_0)} f(x, y) = L.$$



In the limit definition, δ is the radius of a disk centered at (x_0, y_0). For all points (x, y) within this disk, the function values $f(x, y)$ lie inside the corresponding interval $(L - \varepsilon, L + \varepsilon)$.

From the formal definition of limits, we can get several consequences.

- (Uniqueness:) If $\lim_{(x,y)\to(x_0,y_0)} f(x,y)$ exists, then it is unique.
- (Algebra of limits:) Limit laws or properties are established by using the formal definition.

Proposition

For every integer $n \geq 0$, we have

$$\lim_{(x,y)\to(x_0,y_0)} x^n = x_0^n, \qquad \lim_{(x,y)\to(x_0,y_0)} y^n = y_0^n.$$

For example,

$$(a) \quad \lim_{(x,y)\to(-1,3)} x^2 = (-1)^2 = 1, \qquad (b) \quad \lim_{(x,y)\to(-1,3)} y^3 = 3^3 = 27.$$

Theorem (Algebra of limits)

If $\lim_{(x,y)\to(x_0,y_0)} f(x,y)$ *and* $\lim_{(x,y)\to(x_0,y_0)} g(x,y)$ *exist, then*

1. $$\lim_{(x,y)\to(x_0,y_0)} (\alpha f(x,y) + \beta g(x,y))$$
$$= \alpha \lim_{(x,y)\to(x_0,y_0)} f(x,y) + \beta \lim_{(x,y)\to(x_0,y_0)} g(x,y),$$
where α *and* β *are constants.*

2. $$\lim_{(x,y)\to(x_0,y_0)} ((f(x,y)\cdot g(x,y))$$
$$= \lim_{(x,y)\to(x_0,y_0)} f(x,y) \cdot \lim_{(x,y)\to(x_0,y_0)} g(x,y)$$

3. $$\lim_{(x,y)\to(x_0,y_0)} \frac{f(x,y)}{g(x,y)} = \frac{\lim_{(x,y)\to(x_0,y_0)} f(x,y)}{\lim_{(x,y)\to(x_0,y_0)} g(x,y)}$$ *(if* $\lim_{(x,y)\to(x_0,y_0)} g(x,y) \neq 0$*).*

For example,

1. $$\lim_{(x,y)\to(x_0,y_0)} \left(x^2+y^3\right) = \left(\lim_{(x,y)\to(x_0,y_0)} x^2\right)+\left(\lim_{(x,y)\to(x_0,y_0)} y^3\right) = x_0^2+y_0^3.$$

2. $$\lim_{(x,y)\to(x_0,y_0)} \left(x^2 \cdot y^3\right) = \left(\lim_{(x,y)\to(x_0,y_0)} x^2\right) \cdot \left(\lim_{(x,y)\to(x_0,y_0)} y^3\right) = x_0^2 y_0^3.$$

• Limit along a path

Suppose C is a path passing through (x_0, y_0).

Considering values $f(x, y)$ as $(x, y) \to (x_0, y_0)$ along the path C, we express by

$$\lim_{\substack{(x,y)\to(x_0,y_0)\\ C}} f(x, y)$$

the limit of $f(x, y)$ along the path C as (x, y) tends to (x_0, y_0).

Example

$$\lim_{\substack{(x,y)\to(0,0)\\ x=0}} \frac{e^x - e^y}{x + y + 1} = \lim_{y\to 0} \frac{1 - e^y}{y + 1} = \frac{0}{1} = 0.$$

From the uniqueness of limit, it follows that if $f(x, y)$ has different limits as (x, y) tends to (x_0, y_0) along different paths, then the $\lim_{(x,y)\to(x_0,y_0)} f(x, y)$ does not exist.

This is recorded in the following result.

Corollary (Limit does not exist)

Suppose C_1 and C_2 are two paths passing through (x_0, y_0). If

$$\lim_{\substack{(x,y)\to(x_0,y_0)\\ C_1}} f(x, y) \neq \lim_{\substack{(x,y)\to(x_0,y_0)\\ C_2}} f(x, y),$$

then $\lim_{(x,y)\to(x_0,y_0)} f(x, y)$ does not exist.

Example

Does the limit $\lim\limits_{(x,y)\to(0,0)} \dfrac{2x^2-y^2}{x^2+y^2}$ exit?

Solution. Consider two different paths: $x=0$ and $y=0$.

- Along $x=0$: For any $(x,y)\neq(0,0)$ along this path, we have

$$\frac{2x^2-y^2}{x^2+y^2} \underbrace{=}_{put\ x=0} \frac{0-y^2}{0+y^2} = \frac{-y^2}{y^2} = -1 \implies \lim_{\substack{(x,y)\to(0,0)\\ x=0}} \frac{2x^2-y^2}{x^2+y^2} = -1.$$

- Along $y=0$: For any $(x,y)\neq(0,0)$ along this path, we have

$$\frac{2x^2-y^2}{x^2+y^2} \underbrace{=}_{put\ y=0} \frac{2x^2-0^2}{x^2+0^2} = \frac{2x^2}{x^2} = 2 \implies \lim_{\substack{(x,y)\to(0,0)\\ y=0}} \frac{2x^2-y^2}{x^2+y^2} = 2.$$

So the limit $\lim\limits_{(x,y)\to(0,0)} \dfrac{2x^2-y^2}{x^2+y^2}$ does not exist.

Question. What happen if

$$\lim_{\substack{(x,y)\to(x_0,y_0)\\ C_1}} f(x,y) \quad = \quad \lim_{\substack{(x,y)\to(x_0,y_0)\\ C_2}} f(x,y),$$

can we conclude that $\lim_{(x,y)\to(x_0,y_0)} f(x,y)$ exist?

The answer is NO, of course. Please provide a “counterexample” (!).